

Thermal degradation of polystyrene wastes in various solvents

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Abstract

Thermal polystyrene (PS) degradation was investigated in this work harnessing a high-pressure autoclave surrounded by a furnace. *n*-pentane, cyclohexane (CH), and toluene were used separately as the simplest representatives of aliphatic, cyclic, and aromatic solvents. One experiment was solventless. Without solvent at 400 °C, waste samples accrued the weight percentages of 32.0, 46.8, and 21.2 for the solid residue and the liquid and gaseous products respectively. Solvent employment is strongly recommended because it almost doubled the liquid yield while substantially reducing the solid residue to less than 5% and making the total conversion exceed 95%. A solvent:PS ratio of around 2:1 appeared to suffice. From 350 to 450 °C, the operating temperature had only a vague effect on this phase distribution. GC analyses identified methane, C2, C3, and C4 in the gaseous products. GC/MS results showed that cyclohexane usage rendered liquid products of ethyl benzene, methyl ethyl benzene, 1-phenyl naphthalene 1,1-bicyclohexyl, 2-cyclohexyl ethyl benzene. Toluene employment as solvent gave ethyl benzene, methyl ethyl benzene, 1-phenyl naphthalene propyl benzene, 1,2-dimethyl benzene. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

Environmental and economic concerns have caused much interest in developing new techniques to solve plastic waste problems. Landfilling, incineration, and

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re-melting require re-disposal while chemical recycling does not. The latter method involves conversion of polymer waste into fuel or other high value products [1]. An example of this approach is thermal degradation, which has been studied by many researchers [2–7].

Complex reactions occur in thermal degradation of polymers dependent on various factors such as the heating rate, pressure, reaction medium (solution and/or gaseous, oxidative, reductive or inert) and reactor geometry. Polymers have high viscosity, which complicates the process by impeding heat and mass transfer [8,9]. Solvent addition helps with that problem and lowers operating temperatures. Further advantages of this approach are higher liquid yields and easier control.

Sato et al. [4] studied thermal degradation of polystyrene using two different solvent groups at temperatures between 300 and 450 °C. Liquid products toluene, ethyl benzene and styrene were obtained. The first solvent group comprised tetralin and 9,10 dihydroanthracen, which are both hydrogen donors. The solvents in the second group were phenol, 2-naphtol, and diphenylamine all with easily removable hydrogen atom. Higher styrene yield was claimed in the second group.

Heavy oil was the medium used in another thermal degradation research [10] on polystyrene, the temperature ranging from 370 to 415 °C. Nitrogen was employed to carry the products which were mainly styrene, toluene, methyl styrene, ethyl benzene and cumene.

Styrene and its oligomers are normally expected from polystyrene thermal degradation. Secondary products such as toluene and ethyl benzene appear in the absence of a solvent [10,11]. As explained by Sato et al. [4], this may be due to secondary reactions because of mass transfer limitations.

This research on PS thermal degradation harnessed an autoclave pressurized with nitrogen in order to compare aliphatic, cyclic, and aromatic solvents using their simplest representatives.

2. Experimental

Fig. 1 depicts the experimental set-up, which is composed of a 1 L stainless steel reactor (autoclave) surrounded by a furnace with a 1.5 kW electrical heater, a temperature control unit, and a Varian 3400 gas chromatography (GC) equipped with a Porapak Q column. The gas exits through a needle valve to the collection bags on the GC line. This valve allows gas sampling. The reactor lid is fitted with a pressure indicator to follow the reaction progress and remain below 150 bar, the safety limit for the autoclave.

No pre-treatment was carried out on the PS waste foams other than cleaning, washing with water and drying. In each run, about 25 g of foam was weighed and placed in the autoclave. The chosen solvent was added to render a given solvent:waste ratio. The reactor was pressurized and purged with nitrogen three times to ensure an oxygen-free medium. After ensuring that no leakage occurred at 5 atm, heating started. A pre-determined degradation temperature was maintained for 2 h before cooling down to room temperature. Typical temperature and pressure curves are shown in Fig. 2.

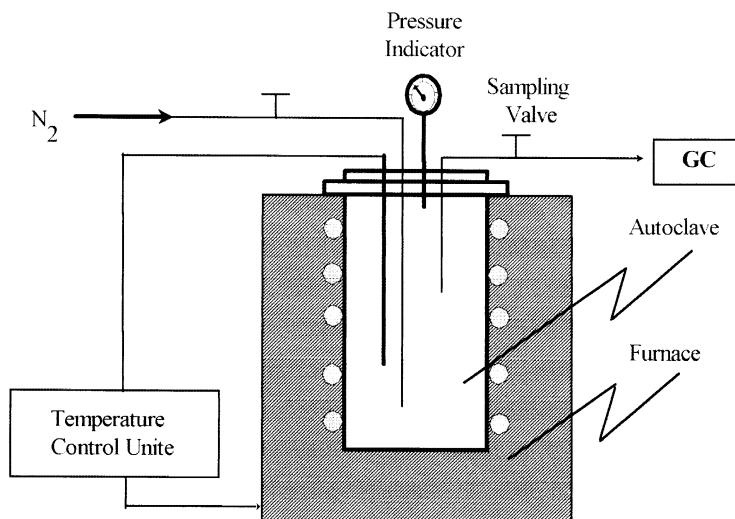


Fig. 1. Simplified schematic representation of experimental system.

The gas phase was emptied to the collection bags on the GC line and analyzed. The solid was filtered from the remains, washed with the same solvent, dried, and then weighed. The washings were added to the filtration liquid to be vacuum distilled to the point of residue decomposition. During distillation, the solvent came

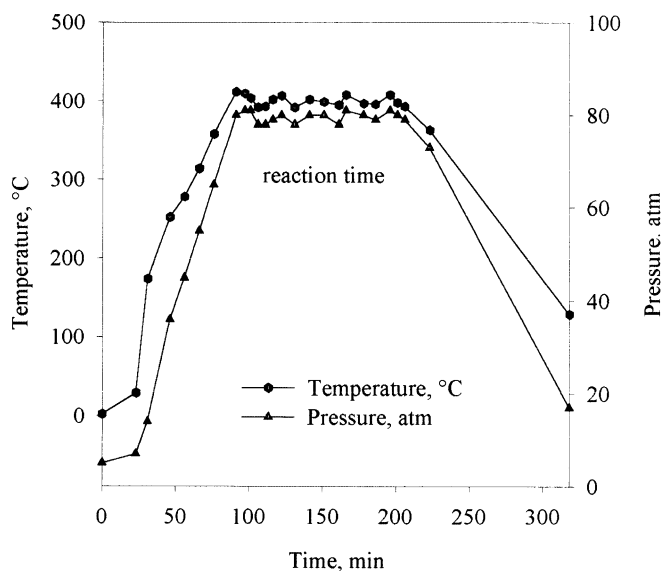


Fig. 2. Typical temperature and pressure curves (at 400 °C, cyclohexane:PS ratio of 4:1).

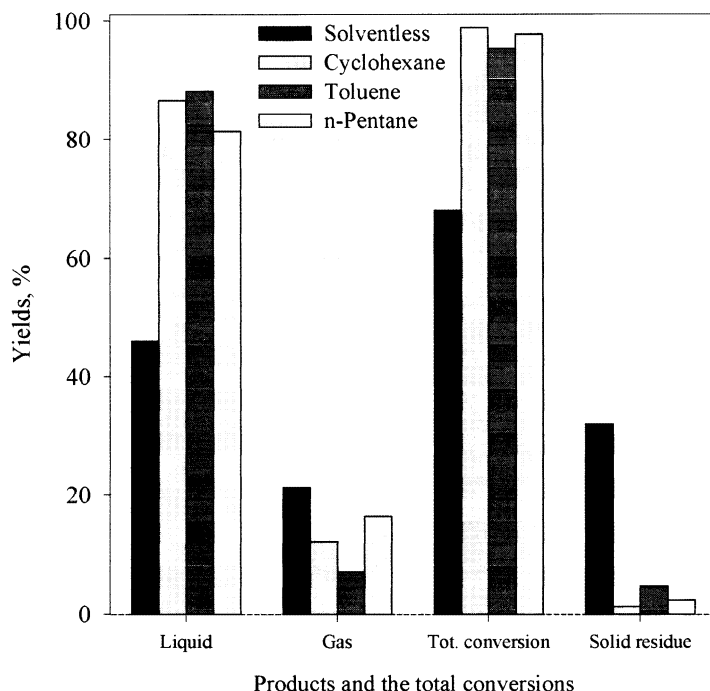


Fig. 3. The effect of solvent type on the gaseous, liquid products and solid residues.

out first as the most volatile component. The subsequent distillates were the light liquid products of the overall process, the viscous liquid residue being oligomers with high molecular weight. The light liquids were analyzed using an HP5890 Plus GC/MS with Ultra-2 column.

The first degradation experiment was carried out at 400 °C without any solvent. This was followed by runs at the same temperature with a solvent:PS ratio of 4:1, using *n*-pentane, cyclohexane, and toluene as the simplest representatives of aliphatic, cyclic, and aromatic solvents. A final set of runs employed cyclohexane only, varying the solvent:PS ratio and the temperature.

The relative mass of the solid residue and the liquid products were calculated. The gas yield was found by difference. The sum of the liquid and gas percentages were defined as the total conversion in the run.

3. Results and discussion

The solventless experiment gave the weight percentages of 32.0, 46.8, and 21.2 for solid, liquid, and gaseous products, respectively, the total conversion, defined above as the sum of the latter two, equaling 68.0%.

As depicted in Fig. 3, any solvent tested here almost doubled the liquid yield while substantially reducing the solid residue to less than 5% and raising the total conversion from 68 to over 95%. This enormous improvement over solventless degradation is probably due to the milder reaction conditions and better heat and mass transfer. Other workers [4,12,13,15,16] have also reported relatively high solid residue and high gas yield in solventless PS degradation (i.e. conventional pyrolysis) compared with that in solution. Fig. 3 displays yield effects of the three solvents. The results are close, cyclohexane showing slight superiority over the other two. Other solvents, particularly those which are hydrogen donors like tetralin and 9,10 dihydroanthracene may affect the yield adversely [4].

The effects of solvent:PS ratio and temperature were investigated in experiments employing cyclohexane. As is clear from Fig. 4, between 2:1 and 10:1, the solvent:PS ratio negligibly altered the yield. For the configuration at hand, a value of 2:1 appeared to be the lowest solvent:PS ratio for which the yield could be calculated with confidence. Carbonized matter was deposited on the reactor walls, below a solvent:PS ratio of about 1:1. Murakata et al. [13] reported a similar difficulty with yield determination. Different geometry and effective stirring may help to reduce this critical value.

The GC analysis of the gaseous output showed methane, C₂, C₃, and C₄ in all cases. Presented in Table 1 are some GC/MS analyses of the light liquid products obtained at 400 °C. Ethyl benzene accrued as reported by other thermal degradation works on polystyrene [4,10]. At solvent ratio of 4:1, toluene generated ethyl benzene, methylethyl benzene, 1-phenyl naphthalene, propyl benzene, 1,2 dimethyl

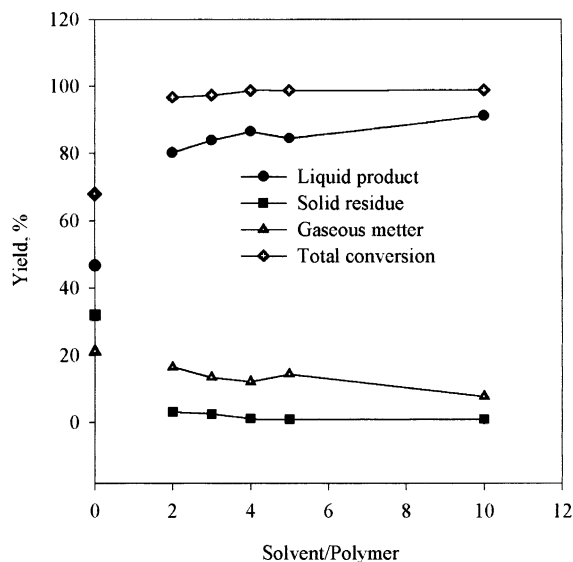


Fig. 4. The effect of solvent:PS ratio on the gaseous and liquid products, solid residue, and the total conversion.

Table 1

Relative amounts of light liquid products at 400 °C (total ion area% of GC/MS chromatograms)

Compound	CH/PS	CH/PS	CH/PS	Toluene/PS
	4/1	5/1	10/1	4/1
Ethyl benzene	13.82	24.97	32.60	36.80
Methyl ethyl benzene	27.33	16.95	15.02	11.65
1-phenyl naphthalene	–	4.30	1.14	1.54
1,1-bicyclohexyl	19.14	5.89	3.80	–
2-cyclohexyl ethyl benzene	18.94	29.29	32.31	–
Propyl benzene	–	–	–	5.21
1,2-dimethylbenzene	–	–	–	1.54
Unidentified	20.77	18.60	15.14	43.26

benzene, and unidentified compounds. Cyclohexane employment at the same solvent:PS ratio apparently caused propyl benzene and 1,2 dimethyl benzene to be replaced by 1,1 bicyclohexyl and 2-cyclohexyl ethyl benzene. The last product indicates that cyclohexane reacted with the light liquid products. Indeed, this product increased with the solvent:PS ratio. Since decomposition took place in nitrogen at 5 atm pressure, the high yields of cyclic products, namely 1,1-bicyclohexyl and 2-cyclohexyl ethylbenzene, can only be explained by reaction with the solvent. The kinetics of this reaction and the overall process may be worth further research since 2-cyclohexyl ethyl benzene is an important raw material. The ethyl benzene content is comparatively much lower in the cyclohexane case and rises back to about the same level only when the solvent:PS ratio is raised to 10:1. The changeover to cyclohexane in contrast more than doubled the methyl ethyl benzene product but the difference diminished with increasing solvent:PS ratio. The solvent to be used may be selected according to the desired product(s).

Since the experimental setup was a closed system, further reactions occurred at the expense of the primary products and caused the styrene monomer to convert into ethylbenzene. Other workers [13,14] using closed systems also reported lack of styrene in the final products. Catalytic hydropyrolysis of PS [14] led to aromatics, ethylbenzene and toluene in particular, accompanied by minor amounts of cyclics and only trace amounts of styrene.

Fig. 5 shows how the yields and the total conversion changed with the operating temperature. The solid residue and the total conversion were apparently unaffected. The gaseous yield slightly rose with the temperature at the expense of the liquid yield, this effect becoming more pronounced at around 450 °C. This was probably due to decomposition of the liquid products to smaller molecules.

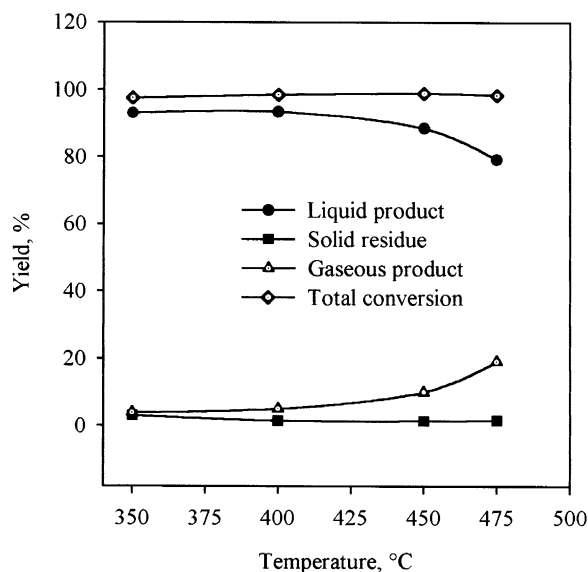


Fig. 5. The effect of temperature on the gaseous and liquid products, solid residue and the total conversion.

4. Conclusions

Thermal degradation of polystyrene waste was investigated as a step in chemical recycling to gain high value products or fuel. The research enlightened many points. Working without solvent at 400 °C gave the weight percentages of 32.0, 46.8, and 21.2 for solid, liquid, and gaseous products, respectively. The total conversion equaled 68.0%. The use of a solvent is strongly recommended as these almost doubled the liquid yield while substantially reducing the solid residue to less than 5% and making the total conversion exceed 95%. Between 2:1 and 10:1, the solvent:PS ratio negligibly altered the yields. Below about 1:1 carbonized matter was deposited on the reactor walls. For the configuration at hand, a value of 2:1 appeared to be the lowest solvent:PS ratio for which the yield could be calculated with confidence. From 350 to 450 °C, the operating temperature had no effect other than slightly increasing the gaseous products at the expense of the light liquids, which probably indicated decomposition.

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